

Fuel Optimal Europa Landing Using Convex MPC

Aayush Iyengar and Owen Willis
Purdue University, West Lafayette, Indiana 47907

Powered descent and landing on Europa presents a unique optimal control challenge due to the low gravity, negligible atmosphere of the moon, and the need to take the minimum amount of propellant to the Jovian environment. This paper presents a fuel-optimal powered descent guidance framework for a Europa lander spacecraft, formulated as a convex Second Order Cone Program (SOCP) with lossless convexification of the nonlinear thrust magnitude constraints as demonstrated by Açıkmeşe and Ploen [1]. The descent problem is solved in three dimensions using a flat Europa approximation with a seven state Cartesian formulation including translational positions, velocities, and vehicle mass. First, an open loop minimum-time feasibility trajectory is computed by bisecting the number of timesteps. This analysis sets the stage for a full horizon minimum-fuel trajectory at the optimal horizon length to then be used as a reference trajectory for any not full horizon solutions. Finally, a receding-horizon Model Predictive Control (MPC) architecture is then implemented to close the loop with true nonlinear RK4-propagated dynamics. The accuracy of the generated solutions from the closed loop system is calculated by adding Gaussian white noise at the initial conditions and by adding in-flight Brownian motion disturbances, and then calculating how well the solutions meet the terminal constraints with a Monte Carlo analysis. The minimum-fuel open loop trajectory converges at $N = 45$ timesteps (90 seconds), consuming 109.7 kg of propellant and achieving a touchdown mass of 409.1 kg at 2.001 m/s. The closed loop MPC controller achieves a terminal lateral position error of 1.68 m and satisfies all velocity constraints, while the Monte Carlo analysis demonstrates that MPC reduces terminal position scatter by approximately two orders of magnitude relative to the open loop baseline under equivalent disturbance levels.

I. Nomenclature

N	Number of discretized timesteps
N_p	MPC prediction horizon length
Δt	Discretization timestep, s
ΔV	Change in Velocity for any Maneuver, m/s
$\mathbf{u}$	Thrust acceleration control vector
$\mathbf{x}$	Full state vector
Γ	Thrust acceleration magnitude, m/s ²
η	Log mass state, $\ln(m)$
m	Vehicle mass, kg
m_0	Initial vehicle mass, kg
g_E	Europa surface gravitational acceleration, m/s ²
I_{sp}	Specific impulse, s
α	Propellant depletion coefficient, $1/(I_{sp}g_0)$
$T_{\min}$	Minimum thrust magnitude, N
$T_{\max}$	Maximum thrust magnitude, N
$\theta_{\max}$	Maximum thrust tilt angle from vertical, deg
γ_{gs}	Minimum glide-slope angle, deg

II. Introduction

Europa is considered one of the most scientifically interesting destinations for future robotic exploration missions due to the presence of a global subsurface ocean on the planet. Europa is a target of astrobiological interest as the surface material might have some life or habitability characteristics that are indicative of the ocean below. Though orbital and flyby missions can be very effective in studying Europa through remote observations, a landing mission will allow the study of the surface material through in-situ measurement techniques. Safe landing of a vehicle on the surface of Europa presents a challenging control problem, mainly because of the following factors: autonomous navigation, stringent terminal constraints, limited propellant, and uncertain initial conditions for the approach phase.

The current work examines the EDL guidance architecture for a Europa lander. The complete EDL problem is decomposed into two distinct stages: the Entry stage (E) and the Descent/Landing stage (DL). The Entry stage controls the craft to a feasible handoff point starting from its Jovian approach trajectory, while the Descent/Landing stage controls the lander from the handoff point to a final target state of landing. However, the E phase was mostly done as a way to get realistic starting conditions for the DL phase. As such, this paper will focus entirely on the dynamics and results of the DL phase, with the initial conditions being derived from the E phase. The staged decomposition is convenient since the two segments of the mission have differing goals, constraints, and guidance characteristics. The Entry stage focuses on placing the spacecraft in a feasible terminal region, while the Descent/Landing stage emphasizes accurate state control and thrust management.

The powered descent portion of the problem is especially well matched for optimization based control methods. In this work, the DL phase is formulated using model predictive control (MPC), where a finite-horizon optimal control problem is repeatedly solved using the current lander state. This allows the controller to update the planned descent trajectory as the vehicle evolves, making the approach more suitable for correcting deviations from the nominal trajectory or from the Entry phase handoff state. The MPC formulation therefore provides a feedback-based extension of traditional open-loop powered descent guidance approaches.

Several previous works influenced the formulation of this project. Açıkmeşe and Ploen developed a convex programming approach for powered descent guidance for a Mars landing. This showed that a minimum-fuel landing problem with thrust constraints can be solved efficiently using convex optimization methods [1]. Their work provides the foundation for optimization based lander guidance. Sostaric and Rea compared powered descent guidance methods for Moon and Mars landing applications, describing the tradeoffs among guidance approaches used in previous planetary landing programs [2]. Their study motivates the need to evaluate guidance algorithms by optimality, practical implementation, robustness, constraint handling, and terminal accuracy. Szmuk, Reynolds, and Açıkmeşe later presented an indirect optimization framework for constrained fuel- and time-optimal 6DOF powered descent guidance, including practical thrust and attitude limitations [3]. This work demonstrates the importance of modeling realistic vehicle constraints in powered descent - mainly for open-loop trajectory generation.

The cited works show that planetary powered descent guidance can be treated as a constrained optimal control problem with realistic constraints. However, most of these studies focus on Moon or Mars landing scenarios and emphasize either open-loop trajectory generation. In contrast, this project applies an MPC-based guidance framework to a Europa lander scenario and explicitly separates the EDL problem into Entry and Descent/Landing phases connected through a handoff state. This structure allows the project to study not only the feasibility of powered descent guidance, but also the sensitivity of the landing phase to the conditions delivered by the Entry phase.

The primary contribution of this project is the development and numerical evaluation of a phased Europa lander EDL guidance architecture. By separating the problem into an Entry phase and an MPC-based Descent/Landing phase, the project provides a structured method for analyzing how handoff conditions affect terminal landing performance. The resulting framework generates feasible landing trajectories, enforces relevant control and terminal constraints, and assesses the ability of the controller to guide the lander to a safe touchdown condition under representative Europa landing assumptions.

III. Project Objectives

Objectives of the project have been classified into three levels based on their technical complexities. At Level 1, an attempt was made to obtain the basic optimal control formulation of the Europa lander system. In Level 2, the objective was to develop a receding horizon MPC controller for the descent and landing of the system. The final stage of Level 3 was to add more realistic (and difficult) functionality to our MPC controller.

A. Level 1 Objectives

The first objective level was to demonstrate a complete understanding of the 3DOF Europa powered descent guidance problem. This required deriving the translational equations of motion, defining the state and control variables, and formulating the relevant constraints. The main constraints considered included thrust magnitude limits, glide-slope constraints, mass bounds, and terminal landing requirements. The continuous time dynamics were then discretized to produce an optimal control problem suitable for numerical solving. The final Level 1 goal was to solve for a full horizon convex reference trajectory, which provided a nominal descent trajectory for future MPC tracking.

B. Level 2 Objectives

The second objective level was to extend the full horizon reference solution into a closed loop framework using convex model predictive control. Rather than applying the full open loop reference trajectory without correction, the MPC controller repeatedly solves a finite horizon optimization problem using the current lander state. This allows the controller to update the commanded thrust profile as the vehicle descends and to correct deviations from the nominal reference. Level 2 therefore included implementation of the receding-horizon MPC controller, development of a closed loop 3DOF simulation with state feedback, and comparison of MPC performance against the original open-loop reference trajectory.

C. Level 3 Objectives

The third objective level originally included several ambitious extensions, such as state estimation, 6DOF dynamics, and sequential convex programming for reference trajectory generation. Within the project scope, the completed Level 3 objective was a Monte Carlo robustness analysis. This study perturbed the initial handoff conditions and evaluated whether the MPC controller could still guide the vehicle to a feasible terminal landing state. The Monte Carlo results were used to assess sensitivity to handoff uncertainty, quantify the spread in terminal position and velocity errors, and evaluate the reliability of the controller across a range of perturbed cases.

D. Summary of Completed Objectives

Overall, the project completed the Level 1 convex powered-descent formulation, the Level 2 closed-loop MPC implementation, and the Monte Carlo robustness portion of Level 3. The remaining Level 3 objectives, including state estimation, 6DOF modeling, and SCP refinement, were identified as future extensions rather than completed work. This progression still provided a complete project story: a nominal optimal control problem was solved, an MPC controller was implemented for close loop solution, and the resulting controller was evaluated under uncertain handoff conditions.

E. Group Member Contributions

Aayush was responsible for the initial linearized 3 DOF setup, reference min-fuel convex trajectory calculation, the initial MPC implementation based on this reference, and the open loop introduction of Brownian motion to the trajectory. Owen set up the RK4 nonlinear dynamics model to be used within the MPC implementation, assisted with MPC implementation, and performed a lot of work with the Monte Carlo robustness simulations with different disturbance levels. Both members worked together extensively on integration, testing, and the analysis of simulated results, and all written and presentation materials.

IV. Proposed Method and Approach

A. Phased EDL Problem Architecture

The Europa entry, descent, and landing problem was separated into two primary phases: the Entry (E) phase and the Descent/Landing (DL) phase. This decomposition was used because the two portions of the trajectory have different objectives and control requirements. The Entry phase is responsible for reducing the vehicle's energy and guiding the lander toward a feasible terminal condition for powered descent. The DL phase then uses this terminal condition as its initial state and guides the vehicle to a safe touchdown using a receding-horizon model predictive controller.

The connection between the two phases is defined through a handoff state. In this project, the handoff state represents the vehicle condition at the end of the Entry phase and the beginning of the DL phase. This state provides the initial

condition for the powered-descent optimization problem and includes the lander's position, velocity, and mass state:

$$\mathbf{x}_{\text{handoff}} = \begin{bmatrix} \mathbf{r}_{\text{handoff}} \\ \mathbf{v}_{\text{handoff}} \\ \mathbf{z}_{\text{handoff}} \end{bmatrix}, \quad \eta = \ln(m). \quad (1)$$

Using a handoff state allows the Entry and DL phases to be solved as separate but connected optimal-control problems. The Entry phase is evaluated based on whether it can deliver the vehicle to an acceptable handoff region, while the DL phase is evaluated based on whether the MPC controller can guide the lander from that handoff state to the desired terminal landing condition. This architecture also provides a natural way to study robustness, since perturbations in the handoff state can be introduced to evaluate the sensitivity of the DL controller to Entry-phase dispersion.

B. Vehicle State, Control, and Coordinate Definitions

The Descent/Landing phase is modeled using a three-degree-of-freedom translational dynamics model. The lander position is defined in a local Cartesian frame fixed to the Europa landing site, where the horizontal coordinates are denoted by x and y , and the vertical altitude coordinate is denoted by z . The position and velocity vectors are therefore

$$\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_x \\ v_y \\ v_h \end{bmatrix}. \quad (2)$$

The full state vector includes position, velocity, and log mass:

$$\mathbf{x} = \begin{bmatrix} x & y & z & v_x & v_y & v_h & \eta \end{bmatrix}^T, \quad \eta = \ln(m). \quad (3)$$

The log-mass state is used because it allows the thrust and mass-coupling constraints to be written in a form more suitable for convex optimization.

The control input is defined using thrust acceleration rather than thrust force directly. Let $\mathbf{T}$ be the physical thrust vector and m be the lander mass. The thrust acceleration vector is

$$\mathbf{a} = \frac{\mathbf{T}}{m} = \begin{bmatrix} a_x \\ a_y \\ a_h \end{bmatrix}. \quad (4)$$

An auxiliary control variable σ is introduced to represent the thrust acceleration magnitude:

$$\sigma = \frac{\|\mathbf{T}\|_2}{m}. \quad (5)$$

The full control vector is then

$$\mathbf{u} = \begin{bmatrix} \mathbf{a}^T & \sigma \end{bmatrix}^T = \begin{bmatrix} a_x & a_y & a_h & \sigma \end{bmatrix}^T. \quad (6)$$

Several simplifying assumptions are made in the 3DOF formulation. The vehicle is treated as a point mass, attitude dynamics are neglected, and thrust is assumed to be directly commandable through the acceleration vector $\mathbf{a}$. Europa's rotation is not included in the dynamics, so Coriolis and Centrifugal acceleration terms are omitted. Gravity is modeled as acting in the negative altitude direction, and the local vertical direction is aligned with the z -axis. Under these assumptions, the model captures the dominant translational motion required for powered descent while remaining simple enough for convex trajectory optimization and receding-horizon MPC implementation.

C. Continuous-Time Equations of Motion

For the optimization problem, the Descent/Landing phase is modeled using 3DOF translational dynamics with constant Europa gravity. The optimization state is

$$\mathbf{x} = \begin{bmatrix} x & y & z & v_x & v_y & v_h & z \end{bmatrix}^T, \quad \eta = \ln(m), \quad (7)$$

and the control vector is

$$\mathbf{u} = \begin{bmatrix} a_x & a_y & a_h & \sigma \end{bmatrix}^\top. \quad (8)$$

Using this notation, the continuous-time dynamics used inside the convex optimization problem are written in state-vector form as

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{v}_x \\ \dot{v}_y \\ \dot{v}_z \\ \dot{\eta} \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \\ v_z \\ a_x \\ a_y \\ a_z - g_E \\ -\alpha\sigma \end{bmatrix}. \quad (9)$$

Here, g_E is the constant gravitational acceleration at Europa's surface,

$$g_E = \frac{\mu_E}{R_E^2}, \quad (10)$$

where μ_E is Europa's gravitational parameter and R_E is Europa's mean radius. The propellant depletion coefficient is

$$\alpha = \frac{1}{I_{sp}g_0}. \quad (11)$$

and is a function of the selected propulsion system of the lander spacecraft. Although constant gravity is used in the convex optimization problem, the closed-loop MPC simulation propagates the lander with altitude-dependent gravity. The truth-propagation state is written as

$$\mathbf{x}_{\text{truth}} = \begin{bmatrix} x & y & z & v_x & v_y & v_h & m \end{bmatrix}^\top. \quad (12)$$

The nonlinear propagation model is

$$\dot{\mathbf{x}}_{\text{truth}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{v}_x \\ \dot{v}_y \\ \dot{v}_z \\ \dot{m} \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \\ v_h \\ a_x \\ a_y \\ a_h - \frac{\mu_E}{(R_E + z)^2} \\ -\alpha m \sqrt{a_x^2 + a_y^2 + a_h^2} \end{bmatrix}. \quad (13)$$

This distinction separates the model used for optimization from the model used for closed-loop validation. The optimizer uses a constant-gravity approximation to preserve a tractable formulation, while the MPC simulation propagates the actual vehicle state with altitude-dependent Europa gravity using numerical integration.

D. Convexification and Path Constraints

Using the same 3DOF translational state as previously defined for the Descent/Landing phase,

$$\mathbf{s} = \begin{bmatrix} x & y & z & v_x & v_y & v_z \end{bmatrix}^\top, \quad (14)$$

where x and y are the horizontal position components and z is altitude above the Europa surface. The translational dynamics are written as

$$\dot{\mathbf{p}} = \mathbf{v}, \quad (15)$$

$$\dot{\mathbf{v}} = \mathbf{u} - g(z)\hat{\mathbf{z}}, \quad (16)$$

where

$$\mathbf{u} = \begin{bmatrix} u_x & u_y & u_z \end{bmatrix}^\top \quad (17)$$

is the thrust acceleration command and $\hat{\mathbf{z}}$ is the local vertical unit vector. Europa gravity as a function of altitude is modeled as

$$g(z) = \frac{\mu_E}{(R_E + z)^2}. \quad (18)$$

The mass state is represented using the log-mass variable

$$\eta = \ln(m), \quad (19)$$

with depletion dynamics

$$\dot{\eta} = -\alpha\Gamma. \quad (20)$$

Here, Γ is an auxiliary thrust-acceleration magnitude variable introduced for convexification. The physical thrust magnitude satisfies

$$\|\mathbf{T}\|_2 = m\|\mathbf{u}\|_2, \quad (21)$$

so the nonconvex thrust constraint can be relaxed using the second-order cone constraint

$$\|\mathbf{u}_j\|_2 \leq \Gamma_j. \quad (22)$$

The acceleration magnitude is then bounded as

$$\sigma_{\min} \leq \Gamma_j \leq \sigma_{\max}, \quad (23)$$

where $\sigma_{\min}$ and $\sigma_{\max}$ are the minimum and maximum allowable thrust acceleration magnitudes. This is the lossless convexification step: the thrust direction and magnitude are separated so that the control magnitude constraint can be written as a convex second-order cone constraint.

The lander is also constrained to remain within an admissible approach cone defined by the glide-slope constraint

$$z_j \geq \tan(\gamma_{\text{gs}}) \left\| \begin{bmatrix} x_j \\ y_j \end{bmatrix} \right\|_2. \quad (24)$$

This prevents the trajectory from approaching the landing site too horizontally and enforces a minimum altitude as a function of horizontal distance from the target.

A thrust-pointing constraint is imposed to limit the lateral component of thrust relative to the vertical component:

$$\left\| \begin{bmatrix} u_{x,j} \\ u_{y,j} \end{bmatrix} \right\|_2 \leq u_{z,j} \tan(\theta_{\max}). \quad (25)$$

This constraint limits the tilt angle of the thrust vector from the local vertical direction. The dry-mass constraint is enforced in log-mass form as

$$\eta_j \geq \ln(m_{\text{dry}}). \quad (26)$$

Using these constraints, the finite-horizon MPC subproblem is written as

$$\min_{\mathbf{s}, \mathbf{u}, \Gamma, \eta} -\eta_{N_h} + \sum_{j=0}^{N_h} (\mathbf{s}_j - \bar{\mathbf{s}}_{k+j})^\top \mathbf{Q} (\mathbf{s}_j - \bar{\mathbf{s}}_{k+j}) \quad (27)$$

$$\text{s.t. } \mathbf{s}_{j+1} = \mathbf{A}_d \mathbf{s}_j + \mathbf{B}_d \mathbf{u}_j + \mathbf{d}_d, \quad (28)$$

$$\eta_{j+1} = \eta_j - \alpha \Gamma_j \Delta t, \quad (29)$$

$$\|\mathbf{u}_j\|_2 \leq \Gamma_j, \quad (30)$$

$$\sigma_{\min} \leq \Gamma_j \leq \sigma_{\max}, \quad (31)$$

$$\eta_j \geq \ln(m_{\text{dry}}), \quad (32)$$

$$z_j \geq \tan(\gamma_{\text{gs}}) \left\| \begin{bmatrix} x_j \\ y_j \end{bmatrix} \right\|_2, \quad (33)$$

$$\left\| \begin{bmatrix} u_{x,j} \\ u_{y,j} \end{bmatrix} \right\|_2 \leq u_{z,j} \tan(\theta_{\max}), \quad (34)$$

$$\mathbf{s}_0 = \mathbf{s}_k, \quad (35)$$

$$\mathbf{s}_{N_h} \in \mathcal{X}_f, \quad N_h = \min(25, N_{\text{rem}}). \quad (36)$$

The objective minimizes fuel consumption through the $-\eta_{N_h}$ term while also penalizing deviation from the reference trajectory $\bar{\mathbf{s}}_{k+j}$. This is done by augmenting the cost function with a quadratic cost of state deviation weighted by the tracking weight matrix $\mathbf{Q}$. $\mathbf{Q}$ is a diagonal matrix whose elements determine how closely a specific state element is tracked. The terminal set $\mathcal{X}_f$ represents the allowable landing region, including bounds on terminal position, altitude, and velocity. The resulting optimization problem remains convex because the dynamics are affine and the thrust magnitude, glide-slope, and pointing constraints are all second-order cone constraints.

E. Discretized Dynamics

The continuous-time dynamics are converted to discrete time so that they can be used in both the full-horizon reference solve and the receding-horizon MPC subproblem. The translational state is

$$\mathbf{s}_k = \begin{bmatrix} x_k & y_k & z_k & v_{x,k} & v_{y,k} & v_{z,k} \end{bmatrix}^\top, \quad (37)$$

and the control input is assumed to be constant over each time interval $[t_k, t_{k+1}]$ under a zero-order hold assumption:

$$\mathbf{u}(t) = \mathbf{u}_k, \quad t \in [t_k, t_{k+1}]. \quad (38)$$

For the optimizer, Europa gravity is treated as a constant disturbance over each discretization interval. This gives the affine discrete-time model

$$\mathbf{s}_{k+1} = \mathbf{A}_d \mathbf{s}_k + \mathbf{B}_d \mathbf{u}_k + \mathbf{d}_d. \quad (39)$$

For a time step Δt , the zero-order-hold matrices are

$$\mathbf{A}_d = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad (40)$$

$$\mathbf{B}_d = \begin{bmatrix} \frac{1}{2}\Delta t^2 & 0 & 0 \\ 0 & \frac{1}{2}\Delta t^2 & 0 \\ 0 & 0 & \frac{1}{2}\Delta t^2 \\ \Delta t & 0 & 0 \\ 0 & \Delta t & 0 \\ 0 & 0 & \Delta t \end{bmatrix}, \quad (41)$$

and

$$\mathbf{d}_d = \begin{bmatrix} 0 \\ 0 \\ -\frac{1}{2}g_E\Delta t^2 \\ 0 \\ 0 \\ -g_E\Delta t \end{bmatrix}. \quad (42)$$

The log-mass state is discretized separately using the auxiliary thrust-acceleration magnitude Γ_k :

$$\eta_{k+1} = \eta_k - \alpha\Gamma_k\Delta t. \quad (43)$$

Thus, the complete discrete prediction model used in the optimizer is

$$\mathbf{s}_{k+1} = A_d\mathbf{s}_k + B_d\mathbf{u}_k + \mathbf{d}_d, \quad (44)$$

$$\eta_{k+1} = \eta_k - \alpha\Gamma_k\Delta t. \quad (45)$$

The discrete model is used as the prediction model inside the convex optimization problem. During closed-loop simulation, however, the first MPC control action is applied to a higher-fidelity truth model with altitude-dependent gravity,

$$g(z) = \frac{\mu_E}{(R_E + z)^2}, \quad (46)$$

and the state is propagated forward numerically. This separates the tractable prediction model used by the optimizer from the nonlinear model used to validate closed-loop performance.

F. Full-Horizon Reference Trajectory Formulation

The full-horizon reference trajectory is calculated by solving the previously defined convex optimal control problem over the entire descent horizon in a single optimization step. Two full-horizon problems are solved in sequence to establish the reference. The first is a minimum-time feasibility problem, where the number of time steps N is treated as the free variable and a bisection search is used to identify the smallest N for which the SOCP remains feasible under the full defined set of constraints. This calculated minimum-time solution serves as a baseline to show that the constraint set is in fact satisfiable and serves as a lower bound on the overall duration for descent. The second full-horizon problem fixes N at the value identified by the bisection search and then looks to minimize the total propellant consumption for descent and landing. The minimum-fuel objective is given by:

$$\min_{\mathbf{s}, \mathbf{u}, \Gamma, \eta} -\eta_N, \quad (47)$$

where $\eta_N = \ln(m_N)$ is the log-mass at the final node $k = N$. This objective is subject to the discretized dynamics, convexified thrust constraints, glide-slope constraint, thrust pointing constraints, log-mass definition, and terminal landing constraints defined in the previous section. The negative log-mass terminal cost $-\eta_N$ is equivalent to minimizing propellant consumption because maximizing the terminal log-mass directly maximizes the remaining vehicle mass at touchdown. Hard terminal constraints are enforced at the final node $k = N$, which require the vehicle to arrive within the specified position tolerance, near the surface, and within the allowable touchdown velocity bounds. The solution to this problem yields the nominal descent trajectory $\bar{\mathbf{s}}(t)$ and thrust profile $\bar{\mathbf{u}}(t)$ to be used as the reference for the future receding-horizon MPC controller.

G. Receding-Horizon MPC Formulation

The receding horizon MPC controller differs from the full horizon reference trajectory problem as it repeatedly solves a finite-horizon convex subproblem at each timestep k using the current lander state as the initial condition. Rather than optimizing over the full remaining trajectory at each step, only a fixed window of $N_h = \min(25, N_{\text{rem}})$ steps is used, where N_{rem} is the number of time steps remaining in the descent. This truncated horizon keeps the solve time per step at a reasonable value, which is independent of how much of the descent has already elapsed. At each step k , the controller solves the SOCP defined in the Convexification and Path Constraints section, with the objective:

$$\min_{\mathbf{s}, \mathbf{u}, \Gamma, \eta} -\eta N_h + \sum_{j=0}^{N_h} (\mathbf{s}_j - \bar{\mathbf{s}}_{k+j})^\top \mathbf{Q} (\mathbf{s}_j - \bar{\mathbf{s}}_{k+j}), \quad (48)$$

where $\mathbf{Q}$ is a diagonal tracking weight matrix that penalizes deviation of the predicted state sequence from the pre-computed minimum-fuel reference trajectory $\bar{\mathbf{s}}_{k+j}$. This combined objective balances fuel minimization with reference tracking, making sure that the MPC solution remains close to the nominal trajectory while still correcting for any deviations that may have been introduced by realistic nonlinear truth dynamics. State feedback enters the MPC framework through the initial condition constraint

$$\mathbf{s}_0 = \mathbf{s}_k, \quad (49)$$

which connects the convex subproblem to the true propagated lander state at the beginning of each prediction window. Only the first control action $\mathbf{u}_0^*$ from the reference is applied to the vehicle, after which the true state is propagated forward one step using the nonlinear RK4 truth model with altitude-dependent Europa gravity. After this step is finished, the problem gets solved again at $k + 1$ with the newly updated state. This receding-horizon structure is what gives the MPC controller its closed-loop character as any deviation from the nominal trajectory introduced by the nonlinear dynamics mismatch, initial condition handoff uncertainty, or in-flight disturbance gets absorbed at the next timestep when the solver gets fed the true state again.

H. Monte Carlo Robustness Analysis

The Monte Carlo robustness analysis evaluates the statistical performance of the open-loop reference trajectory under simultaneous initial condition uncertainty and in-flight disturbances. The analysis is intentionally open-loop, meaning that the control profile from the reference trajectory $\bar{\mathbf{u}}(t)$ gets applied (without any modifications) across all trials, and the SOCP is only solved once throughout the noise propagation. This isolates the effect of disturbances on the trajectory in the absence of feedback, providing a baseline with which the closed-loop MPC performance can be compared.

For each of the $N_{MC} = 50$ trials, the handoff initial condition is sampled from a Gaussian distribution centered at the nominal reference state $\bar{\mathbf{s}}_0$:

$$\mathbf{s}_0^{(i)} \sim \mathcal{N}(\bar{\mathbf{s}}_0, P_0), \quad P_0 = \text{diag}(\sigma_{\text{pos}}^2 \mathbf{I}_3, \sigma_{\text{vel}}^2 \mathbf{I}_3), \quad (50)$$

where $\sigma_{\text{pos}} = 10^{-2}$ and $\sigma_{\text{vel}} = 10^{-2}$ are specified in non-dimensional units and scaled to physical units using the length scale $L^* = z_0 = 5000$ m and velocity scale $V^* = \sqrt{L^* g_E}$. In-flight disturbances are modeled as additive Brownian motion acting only on the velocity states, consistent with the physical interpretation that unmodeled forces such as thruster misalignment and gravity anomalies show up as acceleration perturbations instead of instantaneous position jumps. The theorized disturbance is applied at each RK4 integration step via an Euler-Maruyama discretization,

$$\mathbf{s}_{k+1} = f(\mathbf{s}_k, \bar{\mathbf{u}}_k) \Delta t + G \Delta \mathbf{w}_k, \quad \Delta \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \Delta t \mathbf{I}_3), \quad (51)$$

where the diffusion matrix $G \in \mathbb{R}^{6 \times 3}$ has nonzero entries only in the velocity rows, with intensity $\sigma_{\text{dist}} = 10^{-3}$ (nondimensional) (which makes sense as Brownian motion disturbances are acting only on the velocity states). Terminal performance for each trial is evaluated by recording the final position, velocity, and mass at $t = t_f$ and comparing against the constraint bounds defined in the problem setup. A trial is considered successful if the terminal lateral position falls within the 50 m radius tolerance and all velocity components satisfy the touchdown bounds. The behavior of all terminal states is then used to characterize the statistical spread in landing accuracy and quantify what fraction of open-loop trials achieve a successful touchdown under the applied disturbance levels.

I. Task Summary

Four tasks will be segmented in the next section to demonstrate the development and validation of a complete convex powered descent guidance framework for a Europa lander. Task 1 establishes the full-horizon minimum-fuel reference trajectory, confirming that a feasible descent is achievable under the defined constraint set, and providing the nominal reference for all future tasks. Task 2 extends this result into a closed loop framework using a receding-horizon MPC controller, satisfying all soft-landing constraints while demonstrating that the tracking weight matrix $\mathbf{Q}$ provides a direct tuning mechanism between reference fidelity and constraint satisfaction. Tasks 3 through 5 then stress test the

controller under increasingly realistic uncertainty models, showing how the closed-loop MPC architecture performs against different deterministic and stochastic handoff perturbations, and how an open-loop system may struggle under the same disturbances.

V. Results and Discussion

A. Task 1: Minimum-Fuel Full-Horizon Trajectory

Task 1 computes a min-fuel full horizon trajectory for the lander. Unlike the receding horizon MPC formulation discussed later, this case solves the entire descent as a single optimization problem, from start to finish. This solution, therefore, serves as a useful reference trajectory for the future MPC controller to track. It also establishes the shortest feasible full horizon trajectory for the selected mission parameters and constraints.

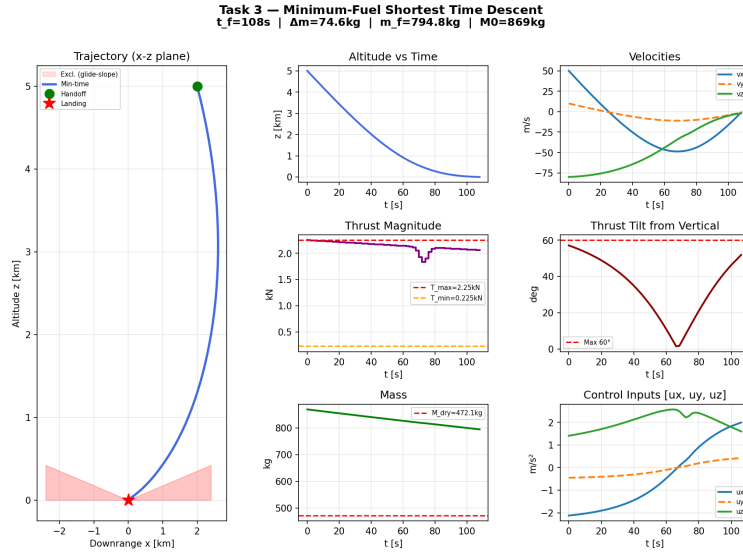


Fig. 1 Minimum-fuel full-horizon result summary

The full horizon solution produced a feasible descent with $N = 54$ time steps and a final time of $t_f = 108$ s. The propellant consumed was approximately $\Delta m = 74.6$ kg, giving a final mass of $m_f = 794.8$ kg. The terminal state was also close to the desired landing condition, with final position errors of $x_f = 0.54$ m, $y_f = 0.12$ m, and $z_f = 0.10$ m away from the origin, and terminal velocities of $v_{x,f} = -0.20$ m/s, $v_{y,f} = -0.20$ m/s, and $v_{z,f} = -2.00$ m/s. These values fall within the target region while satisfying the defined soft landing constraint.

An important result from this case is that a min-fuel solution was found for a 108 s descent, while shorter burn times were found not feasible for the same problem formulation. This suggests that the vehicle is near its performance limit for this initial condition and thrust capability. This is why the thrust magnitude remains near its maximum value for most of the trajectory. This behavior is expected because, at the minimum feasible descent time, the optimizer has limited freedom to just coast. The lander must use nearly all available control to address the initial velocity and reach the terminal landing constraints.

The control history also shows physically reasonable behavior. The vertical acceleration control u_z remains larger than the horizontal components u_x and u_y for most of the descent because the vehicle must counteract gravity and reduce the downward velocity before touchdown. Meanwhile, the horizontal control components change sign during flight. This indicates that the optimizer first redirects the lander toward the target and then reverses acceleration to remove the velocity for the soft landing constraint. The thrust tilt angle from vertical remains smooth and within feasible limits, suggesting that the required thrust-vector pointing behavior is not excessively aggressive. Therefore, although the thrust magnitude is nearly saturated, the commanded thrust direction remains consistent with a physically realistic descent profile.

This solution also serves as the reference trajectory for the later MPC studies. Because this trajectory satisfies the landing constraints while minimizing propellant, it provides a dynamically reasonable baseline for the MPC controller

to track. In the MPC formulation, deviations from this reference are penalized through the state tracking weight matrix (appended to cost function), allowing the controller to remain close to the reference while still re-optimizing in response to nonlinear dynamics, handoff errors, and disturbances.

B. Task 2: Closed-Loop MPC Performance

Task 2 evaluates the nominal closed-loop performance of the receding-horizon MPC controller. The controller uses the previously generated open-loop trajectory as a reference and repeatedly solves a finite-horizon convex optimization problem using the current vehicle state as the initial condition. This allows the lander to track the nominal descent profile while correcting deviations caused by model mismatch and nonlinear dynamics. The tracking matrix used for the augmented cost function for results in Figure 2 and Figure 3 below was: $Q = \text{diag}[10^{-3}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 10^{-2}, 5 \times 10^{-2}]$. Below sections will discuss the differing results with a changing Q .

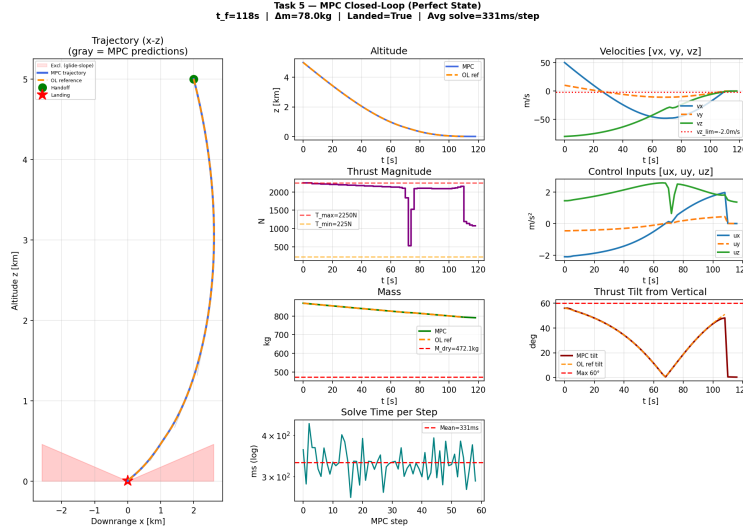


Fig. 2 MPC Closed Loop Result Summary

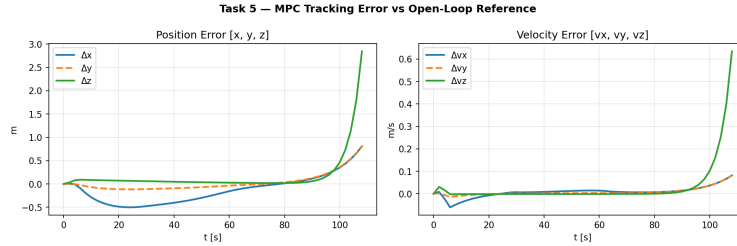


Fig. 3 MPC Reference Tracking Error

The MPC simulation reached the target region with a final horizontal position of approximately $x_f = 1.68$ m and $y_f = 0.33$ m, while maintaining a final vertical velocity of approximately $v_{z,f} = -2.00$ m/s. The fuel consumed was $m_{prop} = 78.0$ kg. These results indicate that the controller was able to satisfy the desired soft landing behavior while remaining close to the reference trajectory. The average solve time was approximately 899 ms per MPC step, which suggests that the optimization problem remained computationally viable for receding horizon simulation.

Figure 2 summarizes the closed loop trajectory, thrust behavior, and terminal performance. The lander follows the reference trajectory while continuously updating the thrust control based on the current state. This feedback structure is important because the nonlinear dynamics are not fully represented by the open loop reference solution. Rather than using a previously solved thrust history, the MPC controller re-optimizes at each step, allowing for deviations to be corrected throughout the descent. The resulting throttle profile also exhibits behavior that is close to bang-bang

control, although it is not perfectly bang-bang due to the tracking objective, path constraints, nonlinear model effects, and receding horizon updates. This is consistent with what is expected from a min-fuel landing problem.

Figure 3 shows the tracking error between the closed loop MPC trajectory and the open loop reference. The largest tracking errors occur in the z direction, which is expected because the primary nonlinear effect included in the model acts in the vertical direction through the gravity term. The tracking error remains bounded for most of the descent, although it increases near the end of the trajectory as the controller prioritizes satisfying the terminal landing constraints over exactly matching the reference. Despite this late increase in tracking error, the soft landing constraints were still met, indicating that the controller maintained the desired terminal behavior. This response is directly influenced by the choice of the reference state tracking weight matrix Q . When a larger Q matrix was used, the maximum position error remained below approximately 3 m, while the maximum velocity error remained near 0.6 m/s. Therefore, increasing the entries of Q forces the MPC solution to more closely track the open-loop reference, while decreasing the entries of Q gives the controller more freedom to deviate for feasibility, disturbance rejection, or propellant efficiency. Figure 4 below shows the result of a less aggressive state tracking weight matrix on tracking error.

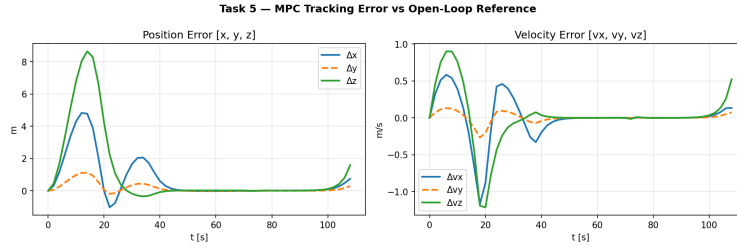


Fig. 4 Tracking Error for Smaller Q

When the tracking weights were decreased to $Q = \text{diag}[10^{-5}, 10^{-5}, 5 \times 10^{-5}, 10^{-4}, 10^{-4}, 5 \times 10^{-4}]$, the MPC controller placed less emphasis on matching the open loop reference trajectory exactly. As a result, the controller was allowed to deviate more from the reference when doing so improved feasibility or helped satisfy the terminal landing constraints. This behavior is expected because smaller entries in Q reduce the penalty associated with position and velocity tracking error, causing the optimizer to prioritize the dynamics, thrust limits, and touchdown constraints more strongly relative to reference tracking. Therefore, the lower Q case can produce larger tracking errors, especially near the end of the descent, but it also demonstrates that the MPC formulation is not simply redoing the open loop reference trajectory. Instead, it actively reoptimizes the trajectory based on the current state and the relative weighting of tracking performance versus constraint satisfaction.

Overall, the Task 2 results show that the MPC controller provides a practical closed-loop extension of the open-loop convex guidance solution. The open-loop trajectory supplies a dynamically reasonable reference, while the MPC formulation adds feedback correction and tunable tracking behavior through the Q matrix. This makes the controller better suited for later robustness studies involving handoff errors, perturbed initial conditions, and stochastic disturbances.

C. Task 3: Initial Condition Robustness Analysis

Task 3 evaluates the robustness of the closed loop MPC controller to errors in the handoff state between the Entry phase and the Descent/Landing phase. This analysis is important because the Descent phase is unlikely to begin exactly from the nominal reference initial condition in a realistic mission scenario. Instead, navigation error, entry dispersions, and other uncertainties would cause the lander to begin powered descent with a perturbed initial state. To evaluate this sensitivity, two types of initial condition perturbation studies were performed: Deterministic perturbations and Gaussian Monte Carlo perturbations.

1. Deterministic Initial Condition Perturbations

The deterministic perturbation study applies fixed one-at-a-time offsets to the nominal handoff state. Position perturbations of ± 100 m were applied independently in x , y , and z , while velocity perturbations of ± 5 m/s were applied independently in v_x , v_y , and v_z . A combined moderate case was also included to test simultaneous perturbations across all state components. For each case, the MPC controller was initialized from the perturbed state and allowed to reoptimize at each time step using the same closed-loop formulation from Task 5.

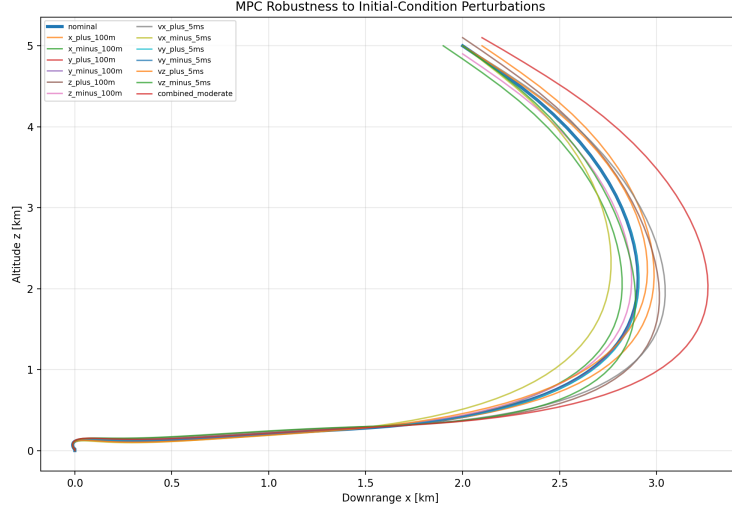


Fig. 5 Deterministic Perturbation Trajectories

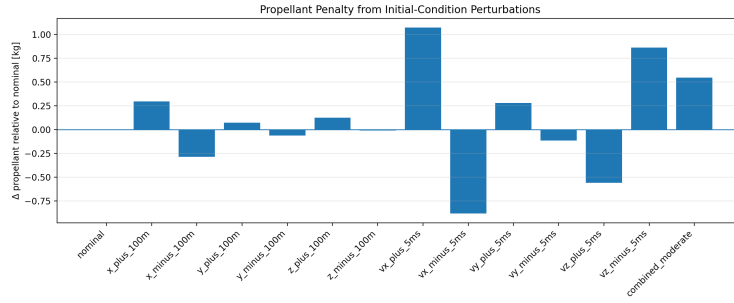


Fig. 6 Difference in Propellant Used from Nominal Case

As seen in Figure 5 deterministic perturbation cases successfully converged to a feasible landing. This result indicates that the MPC controller is able to correct moderate handoff errors in both position and velocity without requiring a major change to the controller formulation. Figure 6 shows that the largest changes in propellant usage occurred for the v_x and v_z perturbation cases, which each changed the propellant used by approximately 1 kg relative to the nominal case. This is physically reasonable because velocity errors directly affect the amount of thrust (and therefore fuel) required to reshape the descent trajectory and satisfy the terminal soft-landing constraints. In contrast, isolated position offsets can often be corrected more gradually over the descent horizon and therefore tend to produce smaller propellant penalties.

2. Gaussian Monte Carlo Perturbations

A Gaussian Monte Carlo study was also performed to evaluate controller performance under randomly distributed handoff errors. The perturbations were sampled using standard deviations of $\sigma_x = \sigma_y = \sigma_z = 100$ m for position and $\sigma_{v_x} = \sigma_{v_y} = \sigma_{v_z} = 5$ m/s for velocity. Unlike the deterministic study, these values represent one-standard-deviation uncertainty levels rather than strict bounds on the sampled perturbations. Each Monte Carlo trial, consisting of 20 trials in total, was propagated using the closed loop MPC controller. Terminal performance, propellant usage, and convergence behavior were then evaluated for both MPC robustness as well as correlation between parameters.

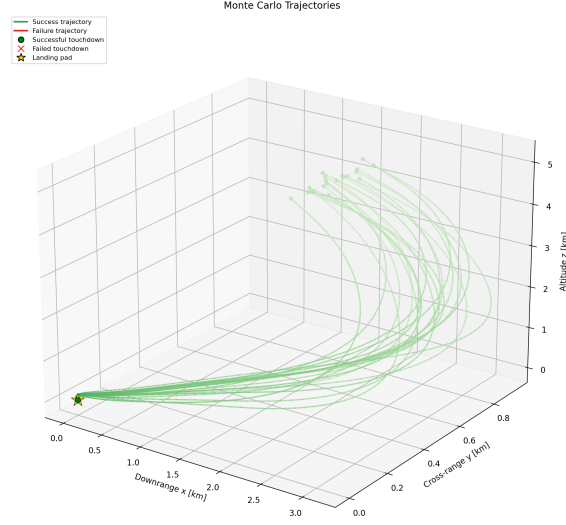


Fig. 7 3D Monte Carlo Trajectories

The Monte Carlo correlation analysis also provides insight into which handoff errors have the most influence on closed loop performance. Figure 8 below shows a correlation heatmap of different simulation parameters. The initial v_x perturbation, $\Delta v_{x,0}$, showed a relatively strong positive correlation above 0.6 with propellant usage, final horizontal position error, and final velocity magnitude. This suggests that downrange velocity error is one of the more important contributors to both fuel cost and terminal accuracy. This matches the result in the deterministic case which further strengthens these findings. This makes sense because an error in v_x changes the horizontal momentum that must be accounted for before touchdown. Correcting this error requires additional acceleration, which can increase propellant usage and affect the final horizontal and velocity errors.

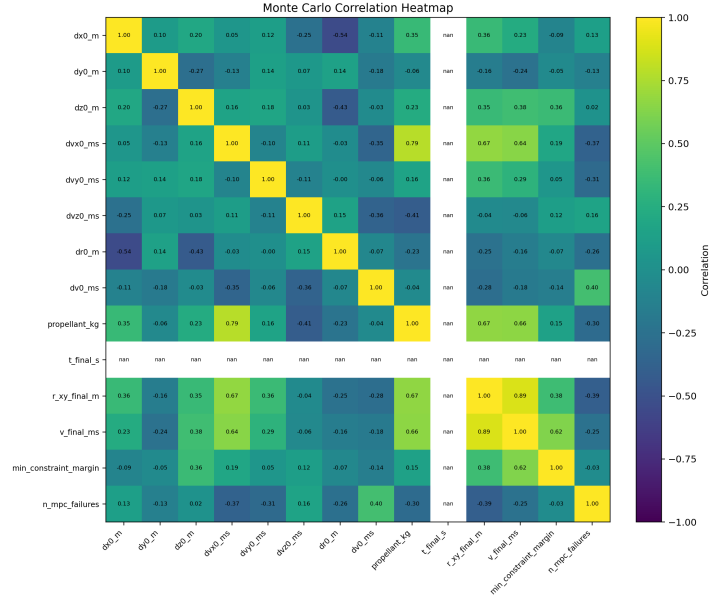


Fig. 8 Parameter Correlation Heatmap

Overall, the initial condition perturbation study shows that the closed loop MPC controller is capable of handling moderate handoff uncertainty. Both the deterministic and Gaussian Monte Carlo cases converged successfully, and the

associated propellant penalty remained small. These results support the use of MPC as a feedback guidance strategy for connecting the entry phase to the descent/landing phase, where exact agreement between the planned and actual handoff states cannot be guaranteed.

D. Task 4 and 5: Sensitivity to Disturbances

Tasks 4 and 5 evaluate how well an open-loop MPC controller is able to control the introduction of initial condition uncertainty coupled with in-flight disturbances throughout the Descent and Landing phases. The open loop reference control sequence $\bar{\mathbf{u}}(t)$, which is just the computed control profile from the minimum-fuel full-horizon solution from Task 1 is applied without modification across all time steps, meaning that no state feedback or trajectory correction maneuvers were applied at any point during the descent of the spacecraft. The two applied noise sources are active simultaneously. First, the initial condition uncertainty was modeled similarly to task 3, where the actual handoff state was sampled from a Gaussian distribution centered at the nominal reference initial condition, with position standard deviation $\sigma_{\text{pos}} = 10^{-2}$ (nondimensional, corresponding to 50 m in physical units) and velocity standard deviation $\sigma_{\text{vel}} = 10^{-2}$ (nondimensional). The in-flight disturbances were modeled as a Brownian motion process acting only on the velocity states (none on position components of the states), with a diffusion intensity of $\sigma_{\text{dist}} = 10^{-3}$ (nondimensional) applied at each RK4 integration step via an Euler-Maruyama discretization. This model for in-flight disturbance serves to simulate any unmodeled anomalies during flight, such as thrust misalignments and additional gravitational perturbations that grow throughout the duration of the descent profile.

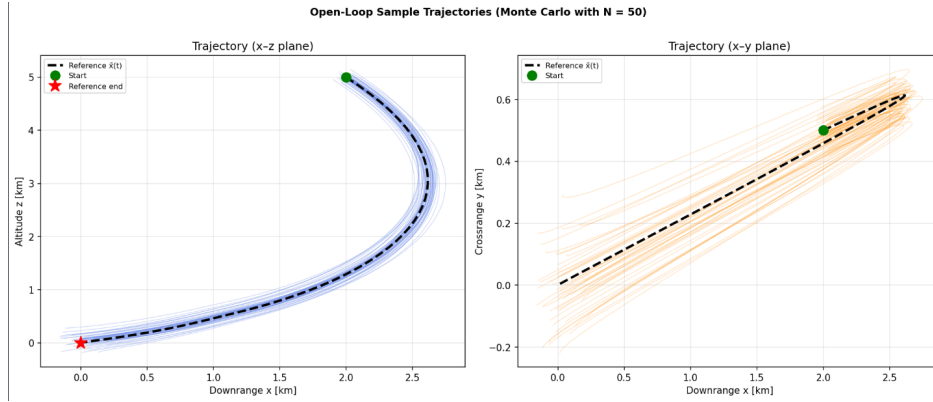


Fig. 9 Open Loop Deviation of Final State with Perturbations of ICs

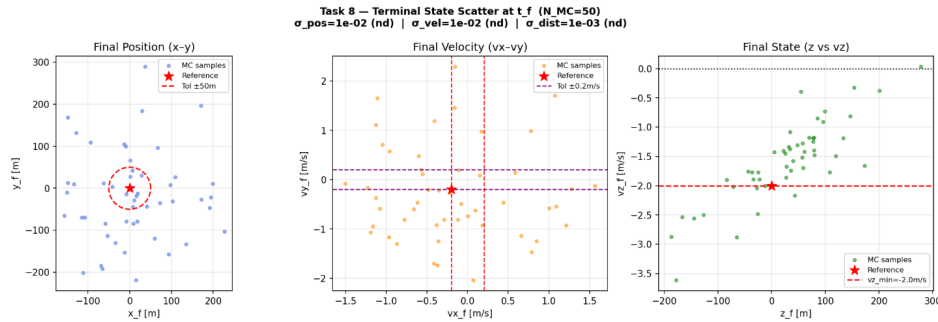


Fig. 10 Final State Dispersion with Gaussian Noise

Figure 9 shows representative noisy trajectories overlaid on the nominal open-loop reference, and it is very clear how the final state begins to significantly deviate from the targeted position and velocity as the disturbances accumulate. Figure 10 shows the Monte Carlo terminal state distribution for both position and velocity components. The terminal scatter results show a few important trends. In the final position plane, the majority of perturbed samples land within approximately ± 200 m of the target site, while only a small number of trajectories end within the 50 m radius landing

tolerance shown by the red dashed circle. This confirms that the open loop controller is not able to reliably satisfy the originally specified landing accuracy constraint under the applied disturbance levels. This can be attributed to the lack of feedback in this simulation, which allows both the initial condition perturbation and the accumulated Brownian velocity increments to propagate freely to the terminal time without correction. Additionally, in the final lateral velocity plane, the scatter showed final states with approximately ± 1.5 m/s, far exceeding the ± 0.2 m/s lateral velocity tolerance originally specified. Again, this is a direct result of the open-loop architecture of this simulation, where a velocity perturbation introduced at $k = 0$ or through Brownian motion at any intermediate step integrates freely across all remaining timesteps, therefore growing linearly with the number of uncorrected steps. The vertical velocity channel is the most robustly satisfied constraint, with nearly all 50 trials producing a final $v_{z,f}$ near the -2.0 m/s reference value. This is physically consistent with the structure of the reference minimum-fuel trajectory, which drives the vehicle to a nearly vertical configuration during the final approach, to make the terminal vertical velocity relatively insensitive to any lateral disturbances. The final spread in altitude z_f across trials is small across all cases, confirming that the open loop burn profile consistently stops the vehicle near the surface regardless of the applied perturbation magnitude.

These open-loop sensitivity results directly motivate the closed loop MPC architecture implemented in Task 2. The biggest limitation that can be identified from this open loop case is that any perturbation, whether from the handoff initial condition or from the introduced in-flight Brownian disturbances, freely accumulates across all the remaining timesteps with no corrective mechanism. On the other hand, the MPC controller is able to address this issue by re-solving the SOCP at each timestep using the true propagated state as the new initial condition, which basically serves to limit the growth of any disturbance to a single 2 second integration interval before the next correction is applied (2 seconds is chosen for feasible on-board computing runtime). The contrast between the two architectures is very clear, as the nominal closed loop MPC result from Task 2 achieved a terminal lateral position error of just 1.68 m and satisfied all velocity constraints, whereas the open loop Monte Carlo scatter placed the majority of trials outside the 50 m position tolerance entirely. This comparison showed the cost of removing feedback as the open loop trajectory is defined as fuel-optimal, but is not able to easily adjust to any deviations from the nominal handoff state. The MPC controller trades a small amount of additional propellant for the closed-loop robustness required to guarantee that the originally defined constraints are satisfied under realistic entry dispersion and in-flight disturbance forces.

E. Modeling Assumptions and Limitations

Several simplifying assumptions were made throughout the descent and landing problem setup that limit how applicable the results presented would be in a flight scenario. The most significant assumption made is the flat-Europa approximation used in the convex optimizer, which replaces the true spherical geometry of Europa with a local Cartesian frame and a constant surface gravitational acceleration of $g_E = 1.315$ m/s², instead of the nonlinear gravitational dynamics that change with altitude. This approximation is well-justified for the short-range, low-altitude descent considered here, but would introduce meaningful error for long range trajectories or potentially higher initial altitude solutions where the curvature of Europa and the altitude dependence of gravity wouldn't be negligible. Another similar assumption made in the problem setup is that Europa's rotation is neglected, which removes the Coriolis and Centrifugal acceleration terms from the equations of motion. For a 90-second powered descent, the rotational contribution is small but would ultimately be non-zero, so for a higher-fidelity simulation, it should definitely be factored into the system's equations of motion. Likewise, the lossless convexification result holds strictly only under these same assumptions for a non-rotating, flat surface with no aerodynamic forces. The results from this simplification need to be reviewed and justified if the problem were extended to include any of these effects, for better practicality.

The descent and landing model is also limited by its 3DOF translational formulation, which treats the vehicle as a point mass and assumes the thrust vector is directly tied to the acceleration control input $\mathbf{u}$. In reality, the thrust direction is coupled to the vehicle attitude, and being able to reorient the thruster requires an angular acceleration that consumes both time and propellant, which is clearly not accounted for in the current model. This leads directly to one of the stretch goals we were unable to implement, a 6DOF extension, which would incorporate rotational equations of motion and enforce attitude rate constraints. Doing so would greatly increase the complexity of the SOCP that needs to be solved, but it would be much more realistic. Additionally, the thrust magnitude bounds were conservatively linearized about the initial mass m_0 instead of the instantaneous mass at every timestep, which introduces another growing approximation error, as in reality, propellant gets consumed over the course of the landing burn. Finally, the open-loop Monte Carlo analysis in Tasks 4 - 5 used a relatively modest disturbance intensity of $\sigma_{\text{dist}} = 10^{-3}$ (nondimensional), and the results should be interpreted within that specific noise regime. Obviously, as higher disturbance levels are introduced, which may be better representative of a degraded thruster or larger navigation uncertainty, more terminal dispersions that

exceed the tolerances demonstrated here would probably be produced.

VI. Conclusion

This project shows the progression from convex powered descent guidance to a robust MPC landing control. First, a min-fuel problem was formulated and solved to generate a feasible reference trajectory. This reference was then implemented with a MPC framework, which allows the controller to reoptimize the trajectory based on current state rather than simply executing a fixed thrust profile. The MPC results showed that reference tracking could be tuned through the tracking weighting matrix Q , with larger weights improving tracking accuracy while smaller weights allowed more freedom for feasibility and disturbance rejection. The controller satisfied the soft landing constraints and produced physically reasonable thrust behavior.

The main takeaway from the robustness analysis is that closed loop feedback is needed for handling uncertainty in the Entry to Descent phase handoff. Both the deterministic perturbation study and the Gaussian Monte Carlo study showed that the MPC controller could correct the handoff errors, including position perturbations on the order of 100 m and velocity perturbations on the order of 5 m/s. The disturbances further showed that while Brownian-like disturbances made terminal constraint satisfaction harder, the MPC controller still performed substantially better than the full horizon open loop execution. Overall, the project completed the Level 1 convex EDL formulation, the Level 2 MPC implementation, and the Monte Carlo robustness portion of Level 3. The remaining Level 3 objectives, including state estimation, 6DOF modeling, and SCP refinement, were identified as future extensions rather than completed work.

A key lesson learned is that solving an optimal trajectory is only the first step. The trajectory must also be analyzed to be usable by a controller under real life uncertainty. Future work in this area should pay particular attention to the connection between Entry phase dispersions and Descent/Landing phase controller authority. Additionally, higher fidelity extensions such as estimator dynamics, quaternion 6DOF motion, and thrust vector pointing limits would be necessary to move the current framework closer to a flight accurate Europa landing scenario.

References

- [1] Açıkmeşe, B., and Ploen, S. R., “Convex Programming Approach to Powered Descent Guidance for Mars Landing,” *Journal of Guidance, Control, and Dynamics*, Vol. 30, No. 5, 2007, pp. 1353–1366. <https://doi.org/10.2514/1.27553>.
- [2] Sostaric, R. R., and Rea, J. R., “Powered Descent Guidance Methods for the Moon and Mars,” *AIAA Guidance, Navigation, and Control Conference and Exhibit*, San Francisco, CA, 2005. <https://doi.org/10.2514/6.2005-6287>, aIAA Paper 2005-6287.
- [3] Szmuk, M., Reynolds, T. P., and Açıkmeşe, B., “Constrained Fuel- and Time-Optimal Six-Degree-of-Freedom Powered Descent Guidance Using Indirect Optimization,” *Journal of Guidance, Control, and Dynamics*, Vol. 43, No. 6, 2020, pp. 1034–1049. <https://doi.org/10.2514/1.G004582>.